The two other readings record the two possible outcomes of the measurement: +1 or −1.

If the apparatus is a quantum system (of course, it must be), then it is described by a space of states. In the simplest description, the apparatus has exactly three states: a blank state and two outcome states. Thus, the basis vectors for the apparatus are

$$|b\}$$

$$|+1\}$$

$$|-1\}.$$

Meanwhile, the basis states of the spin can be taken to be the usual *up* and *down* states:

$$|u\rangle$$

$$|d\rangle.$$

From these two sets of basis vectors, we can build up a composite (tensor product) space of states that has the six basis vectors